

LE MANS ARNAGE AP, France

WMO: 072350

Lat: 47.9458N

Lon: 0.1942E

Elev: 194

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.9	26.8	17.3	13.2	28.6	21.1	15.9	29.8	20.6	50.2	18.6	49.5	4.4	30	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	89.4	68.4	85.4	67.0	81.9	65.7	70.6	84.2	68.8	81.3	67.2	78.4	7.2	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	97.6	74.7	64.6	91.8	73.6	63.0	86.6	71.6	34.5	84.6	33.0	81.3	31.7	78.6	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.3	15.3	13.6	DB	18.1	95.2	5.1	3.5	14.4	97.7	11.4	99.8	8.5	101.7	4.7	104.3	(4)
(5)			WB	17.4	73.9	5.1	2.2	13.7	75.5	10.7	76.8	7.8	78.0	4.0	79.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.1	41.4	42.6	47.2	51.8	58.2	64.5	68.2	67.6	61.7	55.3	47.1	42.3		
	DBStd	11.39	7.20	6.68	5.83	5.87	5.93	6.17	5.65	5.63	5.25	6.05	6.24	7.17		
	HDD50	1059	272	214	125	45	5	0	0	0	0	22	125	251		
	HDD65	4344	730	627	553	396	224	82	30	32	127	302	538	703		
	CDD50	2544	7	7	37	100	259	436	564	547	351	187	37	12		
	CDD65	354	0	0	0	1	13	67	129	114	27	3	0	0		
	CDH74	3498	0	0	0	34	161	690	1229	1097	268	19	0	0		
	CDH80	1150	0	0	0	2	25	221	428	409	63	2	0	0		
	WSAvg	6.4	6.8	7.0	7.0	6.9	6.8	6.6	6.3	5.6	5.7	5.6	5.9	6.7		
	PrecAvg	26.9	2.6	2.3	2.4	2.0	2.3	1.8	2.0	1.8	2.2	2.4	2.8	2.5		
(15) Precipitation	PrecMax	37.7	5.0	4.5	4.0	5.5	5.4	4.3	4.8	5.7	4.6	8.3	6.0	5.4		
	PrecMin	18.3	0.8	0.3	0.1	0.1	0.5	0.0	0.5	0.2	0.4	0.2	0.5	0.4		
	PrecStd	4.3	1.1	1.1	1.1	1.2	1.3	1.0	1.1	1.3	1.3	1.8	1.5	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	62.3	70.4	78.3	82.8	92.8	94.0	95.6	86.2	76.2	64.2	58.2		
		MCWB	54.1	52.3	55.0	60.2	65.1	70.1	69.7	71.0	66.0	63.8	58.4	54.7		
	2%	DB	55.5	57.0	64.7	72.8	78.7	86.0	88.9	89.4	80.9	71.4	60.7	56.4		
		MCWB	52.9	50.5	53.1	58.0	62.6	67.7	68.1	68.7	64.8	62.0	56.6	53.6		
	5%	DB	53.6	54.1	60.6	68.0	74.5	81.6	85.0	84.5	76.6	67.5	58.4	54.6		
		MCWB	51.0	49.8	52.1	55.6	61.0	66.0	66.5	67.2	63.2	60.2	55.0	52.2		
	10%	DB	51.3	52.2	57.4	63.4	70.5	77.3	81.1	80.0	72.8	64.5	56.3	52.8		
		MCWB	48.8	48.8	50.9	53.0	59.1	63.8	65.5	65.3	61.1	59.0	53.1	50.5		
	0.4%	WB	55.2	54.2	57.3	62.6	67.3	72.6	72.7	73.5	69.0	65.5	60.1	56.1		
		MCDB	56.5	58.5	65.9	75.3	79.2	86.9	87.5	90.3	80.0	72.4	62.4	57.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	53.4	52.2	55.1	59.2	64.7	69.3	70.0	70.8	66.5	63.3	57.4	54.2		
		MCDB	55.0	55.3	60.8	70.2	75.2	82.1	84.2	84.3	76.7	68.7	59.9	55.8		
	5%	WB	51.2	50.9	53.5	56.6	62.3	67.2	68.3	68.8	64.6	61.7	55.4	52.6		
		MCDB	53.2	53.5	58.6	65.7	71.6	78.7	81.1	80.8	74.3	66.0	57.9	54.4		
	10%	WB	49.1	49.2	51.8	54.3	60.1	65.1	66.6	66.9	62.8	60.0	53.6	50.8		
		MCDB	51.0	51.8	56.5	61.5	68.9	75.3	78.0	77.0	70.1	63.8	56.0	52.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	13.3	16.4	18.6	18.3	19.3	19.8	20.4	19.5	15.6	12.2	10.6		
		MCDBR	11.0	15.7	22.4	26.4	24.7	25.6	26.1	28.0	25.9	19.4	13.3	10.3		
		MCWBR	9.0	11.0	12.9	13.3	11.7	10.8	9.5	10.7	11.8	10.9	9.6	8.5		
		MCDBR	10.6	13.6	17.6	23.1	22.1	22.6	23.1	23.2	21.7	16.4	11.7	9.8		
	5% WB	MCWBR	9.2	10.5	11.1	12.2	11.2	10.9	9.6	10.0	11.1	10.2	9.3	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.343	0.373	0.397	0.401	0.409	0.403	0.392	0.368	0.362	0.334	0.319		
	taud		2.431	2.379	2.294	2.232	2.266	2.282	2.296	2.346	2.391	2.414	2.436	2.408		
	Ebn at Noon		237	257	266	270	273	271	271	269	265	246	230	222		
	Edh at Noon		21	28	35	41	41	41	40	36	31	26	21	19		
(44) All-Sky Solar Radiation	RadAvg		319	579	946	1400	1651	1845	1768	1550	1209	709	389	273		
	RadStd		30	74	116	162	144	182	118	142	89	59	38	36		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page